

BARENTSBURG, Svalbard and Jan Mayen

WMO: 201070

Lat: **78.056N**Lon: **14.221E**Elev: **76**StdP: **100.42**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-23.6	-21.5	-27.4	0.3	-23.2	-25.3	0.4	-20.7	15.0	-1.6	12.8	-3.4	1.9	40	0.904

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.8	11.0	8.0	9.6	7.1	8.5	6.5	8.3	10.3	7.5	9.2	6.8	8.3	2.8	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
7.1	6.3	8.7	6.3	6.0	8.0	5.7	5.7	7.5	25.5	10.4	23.9	9.2	22.5	8.4	12.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	9.9	8.5	-22.9	13.5	3.4	2.0	-25.3	14.9	-27.3	16.1	-29.3	17.3	-31.7	18.7		
(5)			-23.3	9.9	3.2	1.2	-25.6	10.7	-27.5	11.4	-29.3	12.0	-31.6	12.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.0	-8.3	-9.2	-10.4	-7.9	-2.0	3.2	6.6	5.6	2.2	-2.9	-5.9	-7.6	
	DBStd	7.29	6.41	6.44	5.62	4.94	2.95	1.95	1.78	1.94	2.69	3.88	4.71	5.40	
	HDD10.0	4753	568	537	633	536	373	204	106	138	234	399	477	547	
	HDD18.3	7792	827	771	891	786	631	454	363	396	484	657	727	805	
	CDD10.0	2	0	0	0	0	0	0	2	1	0	0	0	0	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	3.5	4.2	4.2	3.9	3.3	3.2	2.8	3.1	2.9	3.3	3.6	4.1	4.1	
Precipitation	PrecAvg	423	45	35	38	26	21	16	27	33	41	46	48	47	
	PrecMax	697	111	77	99	75	60	64	70	89	86	136	142	123	
	PrecMin	251	3	1	4	1	7	2	4	10	9	4	17	11	
	PrecStd	80	30	25	26	19	12	13	18	19	23	30	31	26	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.3	3.2	2.2	3.0	4.0	8.8	13.5	12.9	8.6	6.2	4.1	3.2	
		MCWB	1.9	1.8	0.9	1.5	2.1	6.0	8.8	8.7	6.6	4.8	2.4	1.9	
	2%	DB	2.3	2.2	1.0	1.8	3.0	7.7	11.7	10.4	7.5	5.1	2.8	1.7	
		MCWB	1.2	1.2	-0.2	0.7	1.5	5.5	8.4	7.6	5.9	3.7	1.3	0.3	
	5%	DB	1.2	0.8	-0.4	0.8	2.2	6.8	10.4	9.0	6.7	3.9	1.7	0.5	
		MCWB	0.1	-0.3	-1.3	-0.3	0.8	4.8	7.6	6.8	5.4	2.7	0.5	-0.6	
	10%	DB	0.0	-0.6	-2.3	-0.6	1.5	6.1	9.3	8.0	5.8	2.6	0.5	-0.5	
		MCWB	-1.1	-1.8	-3.2	-1.6	0.3	4.2	7.0	6.3	4.7	1.5	-0.7	-1.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	2.0	1.2	1.9	2.6	6.7	10.0	9.3	7.2	5.2	2.8	2.2
			MCDB	3.0	2.8	1.9	2.4	3.4	8.3	12.2	12.0	8.1	5.9	3.9	3.1
2%		WB	1.3	1.3	0.0	1.0	1.8	5.7	8.7	7.9	6.3	4.1	1.5	0.5	
		MCDB	2.1	2.1	0.9	1.9	2.6	7.4	10.8	9.6	7.2	4.9	2.5	1.3	
5%		WB	0.2	-0.2	-1.4	-0.2	1.1	5.1	8.0	7.2	5.6	2.7	0.5	-0.4	
		MCDB	1.0	0.6	-0.5	0.7	1.9	6.7	9.9	8.5	6.5	3.8	1.6	0.4	
10%		WB	-1.1	-1.7	-3.1	-1.6	0.4	4.3	7.2	6.6	4.8	1.6	-0.6	-1.5	
		MCDB	0.0	-0.7	-2.3	-0.6	1.3	5.8	9.0	7.9	5.7	2.6	0.4	-0.4	
Mean Daily Temperature Range	5% DB	MDBR	3.9	4.2	4.0	3.7	2.7	2.2	2.8	2.7	2.6	3.0	3.6	3.9	
		MCDBR	4.2	4.8	5.3	3.7	2.9	3.2	4.5	3.9	3.4	3.7	4.0	4.8	
		MCWBR	3.9	4.7	5.3	3.4	2.4	2.3	2.8	2.5	2.9	3.2	3.5	4.6	
	5% WB	MCDDBR	4.3	5.2	5.6	3.7	2.6	3.2	4.0	3.5	3.2	3.5	4.2	5.1	
		MCWBR	4.1	5.2	5.6	3.5	2.4	2.4	2.6	2.4	3.0	3.3	3.8	5.0	
Clear-Sky Solar Irradiance	taub	N/A	0.188	0.248	0.285	0.302	0.296	0.301	0.290	0.246	0.176	N/A	N/A		
	taud	N/A	1.918	2.101	2.245	2.288	2.402	2.468	2.463	2.374	1.966	N/A	N/A		
	Ebn at Noon	0	41	569	770	826	857	835	777	626	31	0	0		
	Edh at Noon	0	5	49	78	91	86	77	64	39	3	0	0		
All-Sky Solar Radiation	RadAvg	0.00	0.03	0.78	2.82	4.73	5.63	4.61	2.92	1.10	0.13	0.00	0.00		
	RadStd	0.00	0.00	0.07	0.34	0.38	0.37	0.33	0.24	0.10	0.01	0.00	0.00		